PAPER_953: Ramanujan-Accelerated S26 Convergence

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** spectral_ladder_26state.py (RamanujanAcceleration) **Calculator:** RamanujanAccelerationCalc (CP4 #537) **CVW:** v2.0.0 compliant

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Abstract

We apply Ramanujan summation techniques to accelerate convergence of the 26-layer buoyancy sum $S_N = \sum_{k=1}^{N} \exp(-[\mathrm{SSq}] \cdot k/26)$. The accelerated form $S_N^{(R)} = S_N + \frac{1}{2}a_N + \sum_{p=1}^{P} \frac{B_{2p}}{(2p)!}\Delta^{2p-1}a_N$ uses Bernoulli numbers B_{2p} and forward differences to improve partial-sum accuracy at low N.

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1. Ramanujan Correction

$$S_N^{(R)} = S_N + \frac{1}{2}a_N + \sum_{p=1}^{P} \frac{B_{2p}}{(2p)!}\Delta^{2p-1}a_N$$

where $a_k = \exp(-[\mathrm{SSq}] \cdot k/26)$ and $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$.

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2. Convergence Comparison

N	S_N	$S_N^{(R)}$
5	partial	accelerated
10	partial	accelerated
26	exact S_{26}	$\approx S_{26}$

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3. Source Data

- **File:** spectral_ladder_26state.py
- **Session:** 214
- **CP4 Class:** RamanujanAccelerationCalc (#537)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Hardy, G.H. (1949) – Divergent Series (Oxford University Press)